

# STA623 - Bayesian Data Analysis

## Practical 4: MCMC with NIMBLE and convergence diagnostics

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**Goals.** Specify and run models in NIMBLE; validate MCMC output against known answers; assess convergence with trace plots,  $\hat{R}$ , bulk/tail ESS and MCSE; diagnose and fix poor mixing. Accompanies Lecture 4.

### Exercise 1: validating NIMBLE against an exact answer

Malaria episodes for 18 children:

```
y <- c(1, 3, 2, 3, 0, 2, 6, 4, 4, 1, 1, 3, 2, 3, 1, 1, 3, 0)
```

- (a) Fit  $Y_i \sim \text{Pois}(\lambda)$ ,  $\lambda \sim \text{Gamma}(5, 2)$  in NIMBLE with 4 chains. Which sampler does NIMBLE choose for  $\lambda$ , and why?
- (b) Compare the posterior mean, 95% equal-tailed interval and 95% HPD interval from the MCMC draws with the exact  $\text{Gamma}(45, 20)$  values. Are differences within Monte Carlo error?
- (c) Now put the prior on  $\theta = \log \lambda$ :  $\theta \sim \mathcal{N}(0, 1.5^2)$ . Which sampler is used now? Compare the posterior of  $\lambda$  with (b).

### Exercise 2: what bad diagnostics look like

Using the model of Exercise 1(c):

- (a) Run 4 chains of only 300 iterations, **no burn-in**, starting from  $\theta = -8, -4, 4, 8$ . Plot the traces and compute  $\hat{R}$  and ESS. Would you trust these results?
- (b) What is the smallest burn-in you would use, judging from the trace plot? Rerun with 2,000 iterations and that burn-in. Are the diagnostics now acceptable?

### Exercise 3: logistic regression that mixes badly

Generate data:

```
set.seed(2026)
N <- 100
age <- runif(N, 20, 60) # age in years; far from 0!
p <- plogis(-6 + 0.12 * age)
seropos <- rbinom(N, 1, p)
```

- (a) Fit logit  $P(Y_i = 1) = \beta_0 + \beta_1 \text{age}_i$  with  $\beta_0 \sim \mathcal{N}(0, 10^2)$ ,  $\beta_1 \sim \mathcal{N}(0, 1)$  in NIMBLE: 4 chains, 3,000 iterations, burn-in 1,000. Compute  $\widehat{R}$ , bulk and tail ESS; look at trace plots, autocorrelation (`coda::autocorr.plot`) and a scatterplot of  $\beta_0$  against  $\beta_1$ .
- (b) Explain the problem. Fix it by centring age (and expressing it per 10 years). Refit and compare diagnostics.
- (c) Alternatively keep the original parameterisation but replace the default samplers with a joint `AF_slice` sampler on  $(\beta_0, \beta_1)$ . Compare.
- (d) From the well-mixing fit, report the posterior median and 95% equal-tailed and HPD intervals for the odds ratio per 10 years of age, and compare with `glm()`.